# 高等数学（下）

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## 偏导数存在、连续、可微三者之间的关 系

$f_x, f_y$ 在点 $P$ 连续

$\Downarrow$ $\not\Uparrow$

$f(x,y)$ 在点 $P$ 可微

$\Downarrow$ $\not\Uparrow$ $\Downarrow$ $\not\Uparrow$

$f(x,y)$ 在点 $P$ 连续 $\rightleftarrows$ $f_x, f_y$ 在点 $P$ 存在 $\rightleftarrows$ $f(x,y)$ 在点 $P$ 方向导数存在

$$\frac{\partial z}{\partial \vec{l}} = \frac{\partial f}{\partial x}\cos\alpha + \frac{\partial f}{\partial y}\cos\beta$$

$$\nabla f(x,y) = \frac{\partial f}{\partial x}\vec{i} + \frac{\partial f}{\partial y}\vec{j}.$$

$$gradf(x,y) = \left\{\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right\}$$

## 一. 链式法则

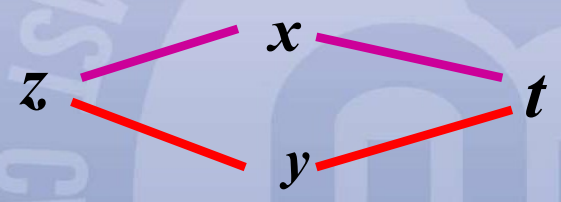
### 一. 全导数

定理 1　设 $x = x(t)$ 和 $y = y(t)$ 在 $t$ 处可微，

而 $z = f(x, y)$ 在相应于 $t$ 的 $(x, y)$ 处可微，

则有 $z = f(x(t), y(t))$ 在 $t$ 可微，且

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial f}{\partial x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y} \cdot \frac{\mathrm{d}y}{\mathrm{d}t}$$



**例** 设 $u = x^2 y + z$，而 $x = t$， $y = e^t$， $z = \sin t$

求全导数 $\dfrac{du}{dt}$.

**解**

$$\frac{du}{dt} = \frac{\partial u}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial u}{\partial y}\cdot\frac{dy}{dt} + \frac{\partial u}{\partial z}\frac{dz}{dt}$$

$$= 2xy + x^2 e^t + \cos t$$

$$= 2te^t + t^2 e^t + \cos t$$

## 二. 偏导数

设 $z = f(x, y)$, 而 $x = \varphi(s,t)$, $y = \psi(s,t)$.

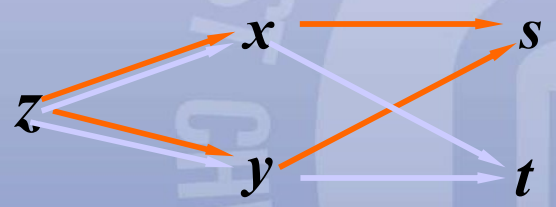
则有 $z = f[\varphi(s,t), \psi(s,t)]$.

如果 $x = \varphi(s,t)$, $y = \psi(s,t)$ 在点 $(s,t)$ 偏导数存在,

且 $z = f(x, y)$ 在对应点 $(x, y)$ 具有连续偏导数.

则复合函数 (2) 在点 $(s,t)$ 的两个偏导数存在,

并且
$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial s}$$
$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial t} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial t}$$



例 设$z=e^u\sin v$，而$u=xy$，$v=x+y$，

求 $\dfrac{\partial z}{\partial x}$和$\dfrac{\partial z}{\partial y}$.

解 $$\frac{\partial z}{\partial x}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial x}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$=e^u\sin v\cdot y+e^u\cos v\cdot 1\ =e^u(y\sin v+\cos v),$$

$$\frac{\partial z}{\partial y}=\frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}$$

$$=e^u\sin v\cdot x+e^u\cos v\cdot 1\ =e^u(x\sin v+\cos v).$$

**例** 设 $z = uv + \sin t$，而 $u = e^t$，$v = \cos t$，

求全导数 $\dfrac{dz}{dt}$.

**解**

$$\frac{dz}{dt} = \frac{\partial z}{\partial u}\cdot\frac{du}{dt} + \frac{\partial z}{\partial v}\cdot\frac{dv}{dt} + \frac{\partial z}{\partial t}$$

$$= ve^t - u\sin t + \cos t$$

$$= e^t\cos t - e^t\sin t + \cos t$$

$$= e^t(\cos t - \sin t) + \cos t.$$

**例** 设$w = f(x+y+z, xyz)$，$f$具有一阶连续偏导数，求$\dfrac{\partial w}{\partial x}$和$\dfrac{\partial w}{\partial y}, \dfrac{\partial w}{\partial z}$.

**解** 令 $u = x+y+z, \quad v = xyz;$

记 $f_1 = \dfrac{\partial f(u,v)}{\partial u}, \quad f_2 = \dfrac{\partial f(u,v)}{\partial v},$

$$\frac{\partial w}{\partial x} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x} = f_1 + yzf_2;$$

$$\frac{\partial w}{\partial y}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y}=f_1+xzf_2;$$

$$\frac{\partial w}{\partial z}=\frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial z}+\frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial z}=f_1+xyf_2;$$

例 设$z=f(x\varphi(xy),x-y,\frac{x}{y})$，求$\frac{\partial z}{\partial x}$

解 设 $u=x\varphi(xy)$，$v=x-y$，$\omega=\frac{x}{y}$，

利用链式法则有

$$\frac{\partial z}{\partial x}=f_1\frac{\partial u}{\partial x}+f_2\frac{\partial v}{\partial x}+f_3\frac{\partial \omega}{\partial x}$$

$$=f_1[\varphi(xy)+x\varphi'(xy)y]+f_2+f_3\frac{1}{y}$$

$$=f_1[\varphi(xy)+xy\varphi'(xy)]+f_2+\frac{1}{y}f_3$$

## 三. 全微分形式的不变性

设 $z = f(x, y)$ 有连续偏导数， 则有

$$\mathrm{d}z = \frac{\partial z}{\partial x}\mathrm{d}x + \frac{\partial z}{\partial y}\mathrm{d}y$$

全微分的运算法则

(1) $\mathrm{d}(cu) = c \cdot \mathrm{d}u$

(2) $\mathrm{d}(u \pm v) = \mathrm{d}u \pm \mathrm{d}v$

(3) $\mathrm{d}(uv) = u \cdot \mathrm{d}v + v \cdot \mathrm{d}u$

(4) $\mathrm{d}\dfrac{u}{v} = \dfrac{v \cdot \mathrm{d}u - u \cdot \mathrm{d}v}{v^2}$

例　设 $u=f(x^2-y^2, xyz, ye^z)$，试求 $\dfrac{\partial u}{\partial x}, \dfrac{\partial u}{\partial y}, \dfrac{\partial u}{\partial z}$ 及 $\mathrm{d}u$.

解　$\mathrm{d}u = f_1 \cdot \mathrm{d}(x^2 - y^2) + f_2 \cdot \mathrm{d}(xyz) + f_3 \cdot \mathrm{d}(ye^z)$

$$= f_1 \cdot (2x\,\mathrm{d}x - 2y\,\mathrm{d}y) + f_2 \cdot (yz\,\mathrm{d}x + xz\,\mathrm{d}y + xy\,\mathrm{d}z)$$

$$+ f_3 \cdot (e^z \mathrm{d}y + ye^z \mathrm{d}z)$$

$$= (2x f_1 + yz f_2)\mathrm{d}x + (-2y f_1 + xz f_2 + e^z f_3)\mathrm{d}y$$

$$+ (xy f_2 + ye^z f_3)\mathrm{d}z$$

则 $$\frac{\partial u}{\partial x} = 2x\, f_1 + yz\, f_2,$$

$$\frac{\partial u}{\partial y} = -2y\, f_1 + xz\, f_2 + e^z f_3,$$

$$\frac{\partial u}{\partial z} = xy\, f_2 + ye^z f_3.$$

例 设 $u = f(xy, xyz)$，求 $u_x$，$u_y$，$u_z$

解 $du = f_1 d(xy) + f_2 d(xyz)$

$$= f_1(ydx + xdy) + f_2(yzdx + xzdy + xydz)$$

$$= (yf_1 + yzf_2)dx + (xf_1 + xzf_2)dy + xyf_2 dz$$

$$\Rightarrow \quad u_x = yf_1 + yzf_2$$

$$u_y = xf_1 + xzf_2$$

$$u_z = xyf_2$$

例 设 $u=f(x,y,z), y=\varphi(x,t), t=\psi(x,z)$ ，其 中 $f,\varphi,\psi$ 均有连续的偏导数，求 $\dfrac{\partial u}{\partial x}$ 和 $\dfrac{\partial u}{\partial z}$.

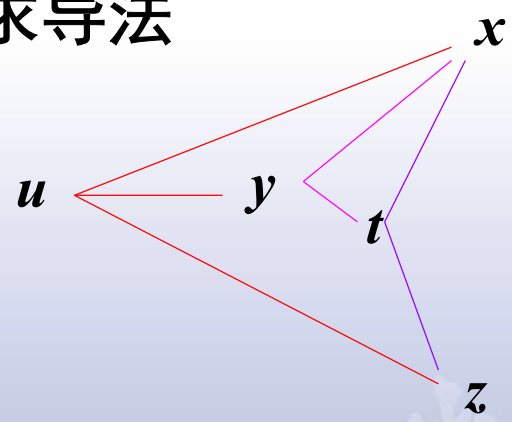
**解法一：利用全微分形式的不变性**

$$
\begin{aligned}
du &= f_x dx + f_y dy + f_z dz \\
&= f_x dx + f_y\left(\varphi_x dx + \varphi_t dt\right) + f_z dz \\
&= f_x dx + f_y\left[\varphi_x dx + \varphi_t\left(\psi_x dx + \psi_z dz\right)\right] + f_z dz \\
&= \left(f_x + f_y\varphi_x + f_y\varphi_t\psi_x\right)dx + \left(f_y\varphi_t\psi_z + f_z\right)dz
\end{aligned}
$$

所以 $\dfrac{\partial u}{\partial x} = f_x + f_y\varphi_x + f_y\varphi_t\psi_x$ $\quad \dfrac{\partial u}{\partial x} = f_y\varphi_t\psi_z + f_z$

## 解法二：利用复合函数求导法

这里函数复合关系为




$$\frac{\partial u}{\partial x} = f_x + f_y \frac{\partial y}{\partial x} + f_y \frac{\partial y}{\partial t}\frac{\partial t}{\partial x}$$

$$= f_x + f_y \frac{\partial \varphi}{\partial x} + f_y \frac{\partial \varphi}{\partial t}\frac{\partial \psi}{\partial x}$$

$$= f_x + f_y \varphi_x + f_y \varphi_t \psi_x$$

$$\frac{\partial u}{\partial z} = f_y \frac{\partial y}{\partial t}\frac{\partial t}{\partial z} + f_z \quad = f_y \varphi_t \psi_z + f_z$$